

ANIF N. SHIKDER

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EXPERIENCE

Machine Learning Researcher

Sep 2022 - Present

Muller Lab (Western University) & Fields Lab for Network Computation (Fields Institute) *Toronto, ON*

- Developed $\mathcal{S}^1$, a **fully interpretable** State Space Model achieving state-of-the-art sequence modeling performance with **full auditability** and **end to end tractability**.
- Created a **mathematical framework** to analyze **latent representations** in sequence models (S4, Mamba), optimizing for long-range dependencies in time-series data.
- Developed **inference-time spectral control methods** to dynamically steer generation in Mamba-based LLMs.

Graduate Teaching Assistant

Sep 2022 - Present

University of Western Ontario *London, ON*

- Taught and mentored students in advanced mathematics (linear algebra, calculus, ODEs/PDEs, and mathematical modeling), translating **complex theoretical concepts into practical problem-solving frameworks** for **150+** students.

Associate Strategy Consultant

Mar 2022 - Sep 2022

Curinos *Toronto, ON*

- Advised **Big Five Canadian and major US banks** on modernizing retail checking suites, streamlining product offerings, and defining competitive **value propositions** across **\$250B+** in deposits.
- Developed a new **advisory product** adopted by **15+** regional banks and fintechs, providing a scalable, data-driven framework to modernize checking and savings portfolios.
- Translated complex financial data into executive-ready insights, directly informing C-suite pricing strategies that optimized margins across **\$50B+** product portfolios.

EDUCATION

University of Western Ontario

Sep 2022 - Aug 2026 (Expected)

PhD in Applied Mathematics, GPA: 4.0/4.0 *London, ON*

Focus: Neural sequence models, mechanistic interpretability, inference-time interventions.

York University

Sep 2020 - Aug 2021

M.A. in Pure Mathematics, GPA: 8.13/9.0 *Toronto, ON*

University of Toronto

Sep 2015 - Aug 2019

HBSc Specializing in Mathematics, GPA: 3.86/4.0 *Toronto, ON*

PUBLICATIONS & RESEARCH

- **Shikder, A. N.**, et al. *Explicit operator explains end-to-end computation in the modern neural networks used for sequence and language modeling*. Manuscript under review, **Physical Review X**.
- Dey, R* and **Shikder, A. N.***, et al. $\mathcal{S}^1$: A Precisely Interpretable State Space Model via an End-to-End Closed-Form Oscillatory Operator. Manuscript under review. **NeurIPS**
- Powanwe, A.*, Liboni, L.*, **Shikder, A. N.**, et al. *Exact expression for maximum Lyapunov exponent during transient dynamics in networks with expressive computational capabilities*. Manuscript under review. **Physical Review Letters**.
- **Shikder, A. N.**, et al. *Complex Walks for Interpretable Associative Memory in State-Space Networks*. Manuscript in preparation.

TECHNICAL SKILLS

Languages & ML:	Python	PyTorch	LangChain	Hugging Face	NumPy
Modeling:	Sequence Models	Representation Learning	Interpretability		
Tools:	Git	Slurm	MATLAB		